

Program Name:

GPA Calculator

Purpose:

A working GPA calculator for a Hunter College undergraduate. The user will be able to enter one or more courses, where each entry would have the program ask whether it should continue. Whenever the user finishes, the program will print the total number of classes and the total semester GPA rounded to the hundredths place.

Inputs:

- A course with at least 2 characters (credits represented by a digit 1-9 and a letter grade given)
- A response to a yes or no to continue adding courses

Outputs:

- Instructions/prompts to display the user's runtime
- A confirmation message that the entry is good
- Error message for invalid inputs
- Final result → number of classes alongside the cumulative GPA that is rounded to the hundredths place (Something like 'Classes taken: 2, Your semester GPA is 3.26')

Variables:

- totalClasses → int → counts how many valid courses were entered by the user
- totalCredits → int → sums up all credits earned
- totalQualityPoints → double → running sum of (credits multiplied by credit points/value)
- userInput → string → stores the raw string the user types in
- continueChoice → string → to store the user's yes/no response
- credit → int → the credit hours taken from the first character of userInput
- grade → string → the grade portion taken from everything after the first character
- points → double → the credit point value matching the grade
- qualityPoints → double → credit multiplied by points for a single course
- cumulativeGPA → double → total quality points divided by the total credits
- rounded → int → cumulativeGPA multiplied by 100 and rounded to nearest whole number
- finalGPA → double → the final GPA rounded to 2 decimal places
- repeat → bool → controls whether the while loop continues

Key Design Choices:

- Being able to input both the credit amount and the grade earned into one string, so that the process is faster for the user
- The code checks the sequence entered is in the correct order: length, credit digit, then the grade— if not then the user gets an error message to retype their entry

- Converting letter cases(upper and lower) in order to make the user's input case-insensitive, so '1a' and '1A' work when entered
- Error message and try again sequence when the user enters an invalid input
- Formula to round the cumulativeGPA $\rightarrow (\text{cumulativeGPA} * 100 + 0.5) / 100.0$
- Checking for 0/empty cases, so the code checks if the totalClasses is 0 and the program will not divide by zero if there were no courses entered

Program Steps:

- 1) Instructions to be read
- 2) While loop begins
- 3) User types an input
- 4) Length of input is validated
- 5) Amount of credits is validated and the character is converted into an integer according to the grade rubric in order to store in 'credit'
- 6) Extract and uppercase the grade $\rightarrow$ everything after the first character is stored in 'grade' $\rightarrow$ Loop through each character and convert to uppercase
- 7) Grade is validated $\rightarrow$ if found then the matching value is stored in 'points'; if not then the 'points' stays at -1 and an error message pops up and you can ask to try inputting again
- 8) Calculating all the data
- 9) Ask if user would like to continue to input more courses $\rightarrow$ if yes, loops back to step 3)
- 10) Display of calculation and displays GPA

Functions:

- 1) toUpperCase
 - a) Takes a string 'word' as an input
 - b) Loops through each character and converts lowercase letters to uppercase
 - c) Returns the string in uppercase
- 2) gradeToPoints
 - a) Takes a string 'grade' as an input
 - b) Uses if/else if in order to match the grade to its point value
 - c) Returns the point value as a double, or -1 if the grade is invalid
- 3) main
 - a) Runs the instructions for the user to read
 - b) Takes in the user's inputs and runs it through a series of validation checks
 - c) Averages the GPA calculation

Sample Input/Output:

- Input:
 - Credits & Grade: 4A
 - Add another course?: Yes/Y or No/N
 - If YES → Repeat from beginning If NO → Enter next course & Grade
- Output:
 - Added 4 credit(s)
 - Grade: A (4 points)
 - Classes Taken: 1
 - Semester GPA: 4.0

Test Cases:

Test #1

- Input: 3B, 3A-, 6C+ [Normal values]
- Expected Result: Added 3 credit(s) x3, Grade: B (3 pts), Grade: A- (3.7 pts), Grade: C+ (2.3 pts)
 - Classes taken: 3, Semester GPA: 2.83

Test #2

- Input: 1FAB [Minimum]
- Expected Result: Added 1 credit(s), Grade: FAB (0 points),
 - Classes taken: 1, Semester GPA: 0.00

Test #3

- Input: 9A+, 9B+, 9C+ [Max]
- Expected Result: Added 9 credit(s) x3, Grade A+ (4 pts), Grade B+ (3.3 pts), Grade C+ (2.3 pts)
 - Classes taken: 3, Semester GPA: 3.2